[1]

No.11 11-12-2024

$$\begin{cases} a\left(\dfrac{\partial^2 w}{\partial x^2}+\dfrac{\partial \psi}{\partial x}\right)=\dfrac{\partial^2 w}{\partial t^2} \\ \dfrac{\partial^2 \psi}{\partial x^2}+\dfrac{12a}{\eta^2}\left(\dfrac{\partial w}{\partial x}-\psi\right)=\dfrac{\partial^2 \psi}{\partial t^2} \end{cases}$$

$u = e^{i\alpha x + \omega t}$ ← ω не имеет отношение с w на верху

$w = c_1 e^{i(\alpha x + \omega t)}$

$\psi = c_2 e^{i(\alpha x + \omega t)}$

$$K^2 + \left(\frac{1}{\alpha} + \frac{\eta^2(1-a)}{12a}\alpha\right)K - \frac{\eta^2}{12} = 0$$

$$\alpha \to 0 \Rightarrow K^2 + \frac{1}{\alpha}K - \frac{\eta^2}{12} = 0$$

$K_1 + K_2 = -\dfrac{1}{\alpha}$ $\qquad K_1 K_2 = \dfrac{-\eta^2}{12}$

$K_1 \to -\infty$, $K_2 \to 0$

$\downarrow$

$\dfrac{A}{B} \to -\infty$ изгебное колебание $\qquad \dfrac{A}{B} \to -\infty$ 

$\dfrac{A}{B} \to 0$ сдвиговые колибание 

МКР: $\dfrac{\partial \psi}{\partial x} = \dfrac{1}{a}\dfrac{\partial^2 w}{\partial t^2} - \dfrac{\partial^2 w}{\partial x^2}$

$$\frac{\partial^3 \psi}{\partial x^3} + \frac{12a}{\eta^2}\left(\frac{\partial^2 w}{\partial x} + \frac{\partial \psi}{\partial x}\right) = \frac{\partial^3 \psi}{\partial t^2}$$

$$\underline{\frac{1}{a}\frac{\partial^2 w}{\partial x^2 \partial t^2}} - \frac{\partial^4 w}{\partial x^4} - \frac{12a}{\eta^2}\left(\cancel{\frac{\partial^2 w}{\partial x^2}} + \frac{1}{a}\frac{\partial^2 w}{\partial t^2} - \cancel{\frac{\partial^2 w}{\partial x^2}}\right) = \frac{1}{a}\frac{\partial^4 w}{\partial t^4} - \underline{\frac{\partial^4 w}{\partial x^2 \partial t^2}}$$

$$\boxed{\frac{\partial^4 w}{\partial x^4} - \left(1+\frac{1}{a}\right)\frac{\partial^4 w}{\partial x^2 \partial t^2} + \frac{12a}{\eta^2}\frac{\partial^2 w}{\partial t^2} + \frac{1}{a}\frac{\partial^4 w}{\partial t^2}}$$ ← вместе 2 ур. получили одно

МКР: $$\begin{cases} a\left(D_{11} w + D_{01} \Psi\right) = D_{tt} w \\ D_{11} \Psi - \dfrac{12a}{\eta^2}\left(D_{01} w + \Psi\right) = D_{tt} \Psi \end{cases}$$

В_РМ: $$\begin{cases} a\left(D_{11} w + D_{01} \Psi\right) = D_{tt} w \\ D_{11} \Psi - \dfrac{12a}{\eta^2}\left(D_{10} w + D_{00} \Psi\right) = D_{tt} \Psi \end{cases}$$

МКЭ: $$\begin{cases} a\left(D_{11} w + D_{01} \Psi\right) = D_{tt} w \\ D_{11} \Psi - \dfrac{12a^*}{\eta^2}\left(D_{01} w + D_{00}^{*} \Psi\right) = D_{tt} \Psi \end{cases}$$

$$D_{11} f_j = \frac{1}{h^2}\left(f_{j+1} - 2 f_j + f_{j-1}\right)$$

$$D_{01} f_j = \frac{1}{2h}\left(f_{j+1} - f_{j-1}\right)$$

$$D_{00} f_j = \frac{1}{4}\left(f_{j-1} + 2 f_j + f_{j+1}\right)$$

$$D_{00}^{*} f_j = \frac{1}{6}\left(f_{j-1} + 4 f_j + f_{j+1}\right)$$

$$D_{tt}^{*} f = \frac{1}{\tau^2}\left(f^{k+1} - 2 f^{k} + f^{k-1}\right)$$

$$D_{01} \Psi = \frac{1}{a} D_{tt} w - D_{11} w$$

МКР → заменим в МКР ур. 2 D_{11} на D_{01} ?

$$\frac{1}{a} D_{tt} D_{11} w - D_{11} D_{11} w - \frac{12a}{\eta^2}\left(D_{10} D_{01} w + \frac{1}{a} D_{tt} w - D_{11} w\right) =$$

$$\frac{1}{a} D_{tt} D_{tt} w - D_{tt} D_{11} w$$

$$\frac{12a}{\eta^2}\left(D_{10} D_{01} w - D_{11} w\right) = 0$$

$$\boxed{\frac{\partial^4 w}{\partial x^4} - \left(1+\frac{1}{a}\right)\frac{\partial^4 w}{\partial x^2 \partial t^2} + \frac{12}{\eta^2}\frac{\partial^2 w}{\partial t^2} + \frac{1}{a}\frac{\partial^4 w}{\partial t^4} = 0}$$

$$D_{10} D_{01} w_i = D_{10}\left(\frac{1}{2h}\left(w_{i+1} - w_{i-1}\right)\right) = \frac{1}{2h}\left(D_{01} w_{i+1} - D_{01} w_{i-1}\right)$$

$$= \frac{1}{2h}\left[\frac{1}{2h}\left(w_{i+2} - w_i\right) - \frac{1}{2h}\left(w_i - w_{i-2}\right)\right]$$

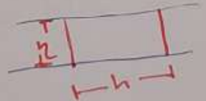
$$= \frac{1}{4h^2}\left(w_{i+2} - 2w_i + w_{i-2}\right)$$

$$D_{01} D_{10} w - D_{11} w = \frac{1}{4h^2}\left(w_{i+2} - 4w_{i+1} + 6w_i - 4w_{i-1} + w_{i-2}\right)$$

$$D_{11} D_{11} w = \frac{1}{h^4}\left(w_{i+2} - 4w_{i+1} + 6w_i - 4w_{i-1} + w_{i-2}\right)$$

$$D_{10} D_{01} w - D_{11} w = \frac{h^2}{4} D_{11} D_{11} w$$

$$\left(2 + 3a\left(\frac{h}{\eta}\right)^2\right) D_{11} D_{11} w - \left(1+\frac{1}{a}\right) D_{11} D_{tt} w + \frac{12}{\eta^2} D_{tt} w + \frac{1}{a} D_{tt} D_{tt} w = 0$$



η – толщина

h – шаг сетки

$$\frac{12a}{\eta^2}\left(D_{10} D_{01} w - D_{11} D_{00} w\right) = O(?)$$

$$D_{11} D_{00} w = \frac{1}{4h^2}\left(w_{i+2} - 2w_i + w_{i-2}\right) = D_{01} D_{10} w$$

$$D_{11} D_{11} w - \left(1+\frac{1}{a}\right) D_{11} D_{tt} w + \frac{12}{\eta^2} D_{tt} D_{00} w + \frac{1}{a} D_{tt} D_{tt} w = 0$$

В $\overrightarrow{\text{РМ}}$

$I,\ \frac{\partial}{\partial x},\ \frac{\partial^2}{\partial x^2} \to I \cdot \frac{\partial^2}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial}{\partial x}$	
$I,\ D_{01},\ D_{11}$	МКР
$D_{00},\ D_{01},\ D_{11}$	ВРМ

$I \cdot D_{11} \neq D_{01} D_{01}$

$D_{00} D_{11} = D_{10} D_{01}$

$D_{11} = d_1^+ d_1^-$

$D_{01} = d_0^+ d_1^- = d_1^+ d_0^-$

$D_{00} = d_0^+ d_0^-$

$$\frac{\partial^2}{\partial x^i \partial x^j} \qquad \frac{\partial^2}{\partial x^k \partial x^l} = \frac{\partial^2}{\partial x^k \partial x^j} \frac{\partial^2}{\partial x^i \partial x^l}$$

$$\frac{\partial^2}{\partial x^i \partial x^j} \frac{\partial}{\partial x^k} = \frac{\partial^2}{\partial x^k \partial x^j} \frac{\partial}{\partial x^i}$$

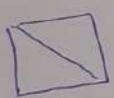
$$\frac{\partial^2}{\partial x^i \partial x^j} = \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j}$$

$D_{ij} D_{kl} = D_{kj} D_{il}$ $\qquad i, j, k, l = 0, 1, 2$

 $\rightarrow D_{ij} D_{kl} = d_i^+ d_j^- d_k^+ d_l^- = d_k^+ d_j^- d_i^+ d_l^-$

$= D_{kj} D_{il}$

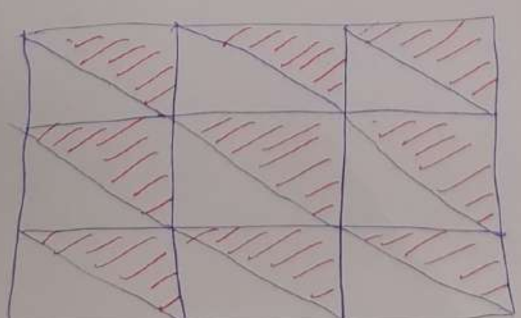
$= D_{il} D_{kj}$



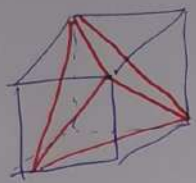
$D^*_{ij} = \frac{1}{2}(D_{ij} + D_{ji})$

$D^*_{ij} D^*_{kl} \neq D^*_{kj} D^*_{il}$

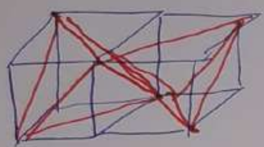
 $\Rightarrow$ $\underset{d_i^+ d_j^-}{\underset{\|}{D_{ij}}} \neq \underset{d_j^+ d_i^-}{\underset{\|}{D_{ji}}}$



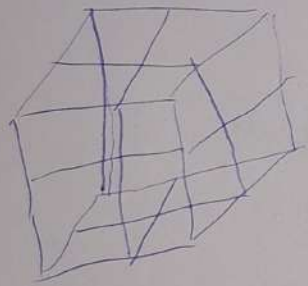
2D

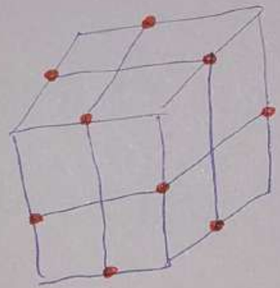


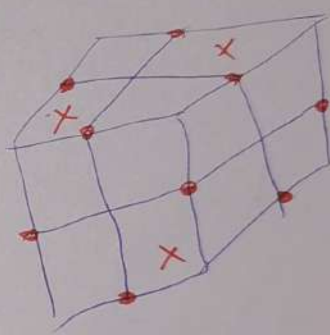
число ячеек уменьшается в 2 раза

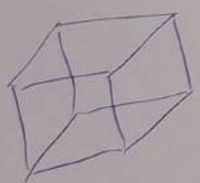


Ажурная схема МКЭ









$d_0^+ \quad d_1^+ \quad d_2^+ \quad d_3^+$

двумерная моментная

$$d_3^+ = \frac{1}{h_3} \begin{matrix} -1 & 1 \\ 1 & -1 \end{matrix} \sim c\,\frac{\partial^2}{\partial x^1 \partial x^2}$$

Ажурная схема в $\mathbb{R}^7$

симплекс в $\mathbb{R}^7 \to$ 8 узлов

проектируем в $\mathbb{R}^3$

симплекс $\longrightarrow$ куб